

Industrial Training

NGCP Nowshera 220 kV Grid Station

Objective

To visit the 220 kV NGCP Nowshera Grid Station, observe real high-voltage equipment, identify major substation apparatus and ratings, and connect classroom power-system concepts with practical grid operation, protection, and safety procedures.

Introduction

A grid substation is a critical junction in the power system where high-voltage electricity is received, switched, protected, transformed, measured, and dispatched toward load centers. The NGCP Nowshera 220 kV Grid Station is an important node in the Khyber Pakhtunkhwa transmission network, receiving power from major sources such as Ghazi Barotha, Mardan, and Sheikh Muhammadi/Peshawar and supplying industrial and regional loads.

During the visit, students observed outdoor switchyard equipment, control-room panels, protection relays, metering systems, synchronizing instruments, and DC backup batteries. The visit emphasized that a modern substation is not only a switching yard; it is a coordinated system for reliable operation and fast fault isolation.

Single Line Diagram

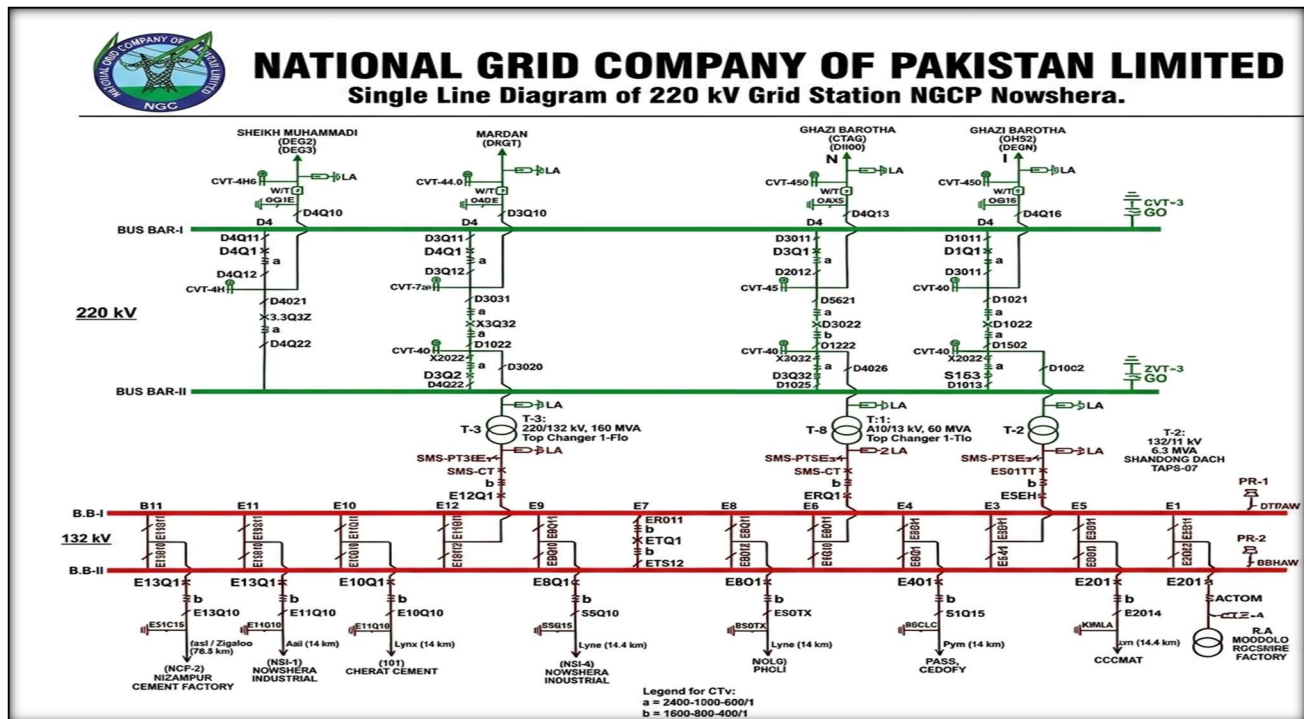


Figure 1. Single Line Diagram of NGCP Nowshera 220 kV / 132 kV Grid Station.

The SLD provides a simplified map of the station. It shows the 220 kV incoming side, 132 kV outgoing side, double bus-bar arrangement, incoming power sources, outgoing feeders, and major switching points.

Surge Arrester



Figure 2. Surge arrester installed in the 220 kV switchyard.

A surge arrester protects expensive substation equipment from lightning and switching overvoltages by conducting surge energy safely to earth.

Parameter	Value
System Voltage	220 kV
Type	Metal oxide, gapless
Height	~3-4 m
Monitor range	0-3 mA

Surge Arrester Monitor



Figure 3. Surge arrester monitor mounted at the base of the arrester.

The monitor measures leakage current through the arrester. Abnormally high leakage can indicate ageing or deterioration of the internal metal-oxide blocks.

Capacitive Voltage Transformer



Figure 4. Capacitive Voltage Transformer used for metering and protection.

A CVT scales high voltage to a safe secondary value for metering, protection relays, and power-line carrier communication.

Parameter	Value
Primary voltage	220 kV
Secondary voltage	110 V typical
Use	Metering, protection, and communication coupling

Disconnectors and Isolators



Figure 5.1. Disconnector / isolator structure in the 220 kV yard.

Disconnectors provide visible isolation for maintenance. They are operated only after the circuit breaker has interrupted current.

Parameter	Value
Voltage class	220 kV
Current rating	1600-2000 A typical
Operation	Motor / manual operation

Disconnectors and Isolators



Figure 5.2. Alternate view of disconnector / isolator arrangement.

The three poles operate together and provide safe separation between energized and de-energized equipment sections.

Transmission Line Towers and Bus Structures



Figure 6.1. Transmission line towers and incoming-line structures.

Transmission towers carry incoming 220 kV conductors, supported by long insulator strings and gantry structures.

Transmission Line Towers and Bus Structures



Figure 6.2. Bus-bar and gantry structures inside the switchyard.

The double bus-bar arrangement improves reliability because feeders can be shifted between buses during operation or maintenance.

Power Transformer



Figure 7. Power transformer / auto-transformer in the grid station.

The auto-transformer steps 220 kV down to 132 kV and is one of the most valuable assets in the substation.

<i>Parameter</i>	<i>Value</i>
Primary voltage	220 kV
Secondary voltage	132 kV
Capacity	250 MVA
Type	Auto-transformer
Cooling	ONAN / ONAF oil cooling

SF6 Circuit Breaker



Figure 8. SF6 gas circuit breaker.

A circuit breaker can interrupt load and fault current. SF6 gas extinguishes the arc when contacts separate.

<i>Parameter</i>	<i>Value</i>
Voltage	220 kV
Breaking current	Up to 40-63 kA typical
Arc medium	SF6 gas
Operating time	~40-60 ms

Local Control Cabinet

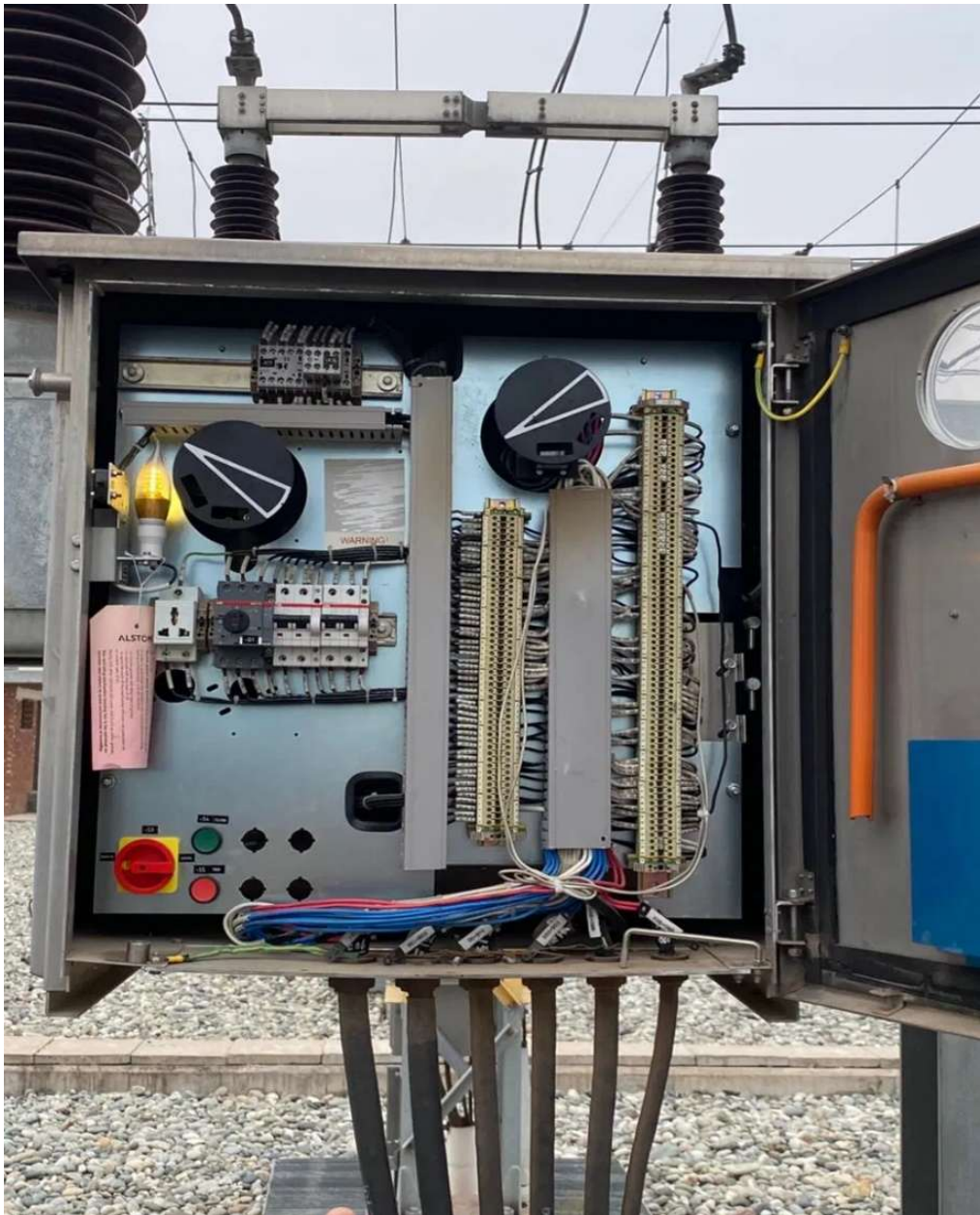


Figure 9. Local Control Cabinet for yard equipment.

The LCC allows local operation, indication, auxiliary wiring termination, and maintenance testing for the associated breaker or isolator.

Ghazi Barotha Line Protection Panel



Figure 10.1. Ghazi Barotha line protection panel.

Line protection relays detect faults and trip the corresponding circuit breaker rapidly to isolate unhealthy sections.

ATR-1 Transformer Protection Panel



Figure 10.2. ATR-1 transformer protection panel.

Transformer protection typically includes differential protection, overcurrent protection, earth-fault protection, Buchholz alarm/trip, and annunciation.

Distance Protection Relay



Figure 10.3. Siemens SIPROTEC 7SA52 distance protection relay.

Distance protection estimates fault location from measured impedance and is a primary protection method for transmission lines.

Main Control Panels



Figure 11. Main control panels / mimic board in the control room.

The control room enables remote monitoring and switching of feeders, bus bars, breakers, and transformers.

Synchronizing Panel



Figure 11.1. Synchronizing panel instruments.

Synchronizing instruments confirm voltage magnitude, frequency, and phase-angle matching before connecting two systems together.

Metering Panel

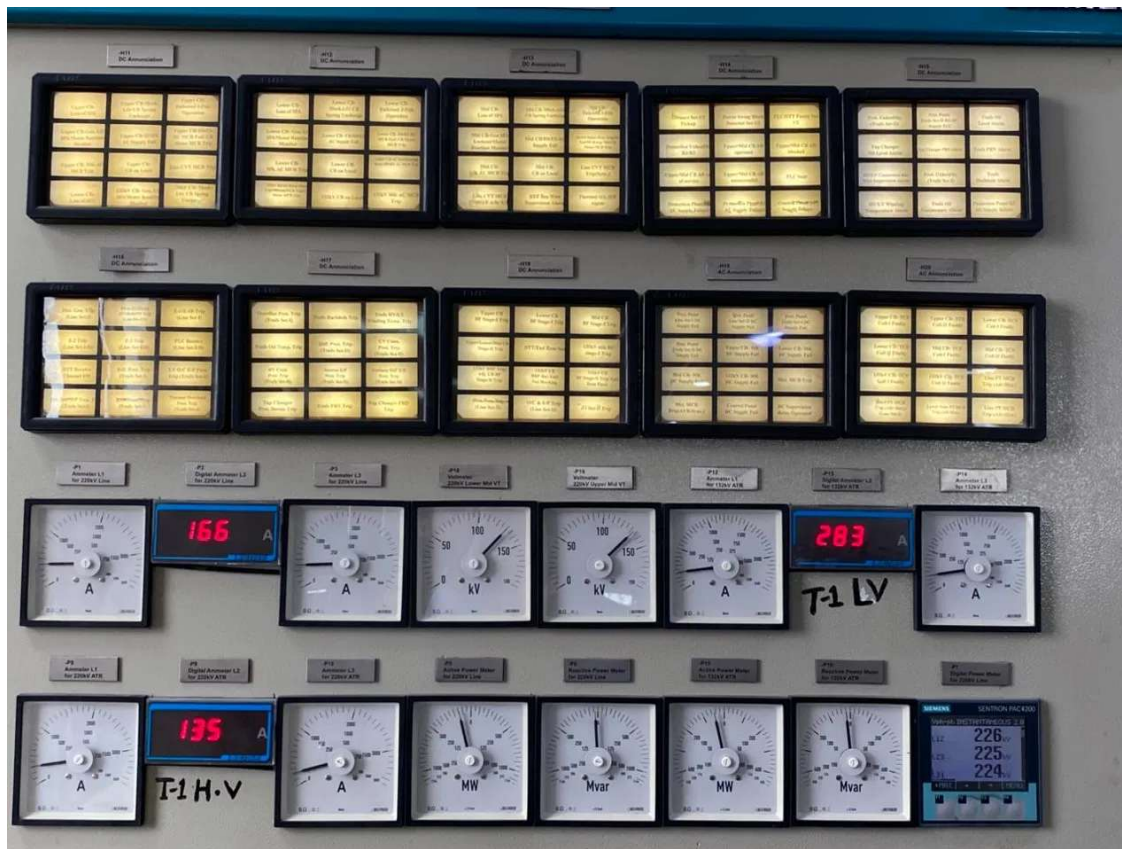


Figure 12. Metering panel with analog and digital measurements.

The metering panel gives operators real-time electrical measurements and alarm indications for system monitoring.

DC Battery Bank



Figure 13. DC battery bank for auxiliary and protection supply.

The battery bank keeps protection and tripping systems alive even if AC auxiliary power is lost.

Parameter	Value
Type	Flooded lead-acid, GFX-300
Per-cell voltage	2 V
Capacity	300 Ah
Total system voltage	~110 V DC
Purpose	Trip coils, relays, and emergency power

Summary of Observed Apparatus

#	Apparatus	Voltage / Rating	Key Role
1	SLD / Wall Diagram	-	Overview map of station
2	Surge Arrester	220 kV	Overvoltage protection
3	Surge Arrester Monitor	0-3 mA	Health check of arrester
4	CVT	220 kV / 110 V	Voltage measurement and carrier coupling
5	Disconnecter / Isolator	220 kV	Visible isolation for maintenance
6	Transmission Towers and Bus Bars	220 kV	Carry and connect power lines
7	Power Transformer / ATR	220/132 kV, 250 MVA	Step-down transformation
8	SF6 Circuit Breaker	220 kV	Fault interruption and switching
9	Local Control Cabinet	-	Local equipment control
10	SIPROTEC Protection Relays	-	Fault detection and breaker tripping
11	Control Room Mimic Board	-	Remote monitoring and control
12	Metering Panel	-	Real-time measurements
13	Synchronizing Panel	-	Safe paralleling of systems
14	DC Battery Bank	110 V DC	Backup power for relays and trips

Key Learning Outcomes

- A 220 kV grid station is a coordinated system of switching, transformation, metering, protection, communication, and backup-power equipment.
- Protection equipment such as surge arresters, relays, circuit breakers, and DC batteries is essential because fault isolation must occur quickly and reliably.
- Instrument transformers such as CVTs allow dangerous high-voltage quantities to be measured safely by low-voltage devices.
- The control room and mimic board allow operators to supervise the entire station without entering the high-voltage yard.
- Substation equipment ratings are selected according to voltage level, current, fault level, insulation clearance, and system reliability requirements.

Safety Observations

- Safe clearances are essential at 220 kV because dangerous flashover can occur without direct contact.
- Outdoor equipment must be operated only by authorized personnel using approved switching procedures.
- Isolators provide visible isolation, but circuit breakers must interrupt load or fault current before isolation.
- The DC battery system is critical because protection and trip circuits must remain functional during AC supply failure.

Conclusion

The visit to the NGCP 220 kV Grid Station at Nowshera provided a valuable practical understanding of high-voltage power-system operation. The observed equipment showed how transmission-level substations receive, switch, transform, protect, measure, and dispatch electrical power. The visit connected classroom concepts such as bus bars, instrument transformers, circuit breakers, protection relays, synchronizing, and backup DC supply with real engineering practice. It also emphasized that reliable grid operation depends not only on power-carrying equipment but also on protection, control, monitoring, and safety systems working together.